Performance Analysis of FPGA Implementation of 4 x 4 Vedic Multiplier using different adder architectures

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***Abstract*— High-speed applications, such as signal processing and image processing, require fast computations, which in turn require high-speed multiplications. Multipliers are the most essential components in signal processing systems and are used for various operations such as filtering, modulation, coding, and decoding. Therefore, the development of high-speed multipliers is crucial for meeting the demands of modern digital signal processing applications. There are various techniques used to design high-speed multipliers, such as parallel multipliers, array multipliers, and carry save multipliers. These techniques help in producing fast and efficient multipliers that are capable of meeting the demands of high speed applications. The multiplier blocks are considered as one of the major hardware blocks causing a bottleneck in terms of power dissipation and delay in most of the fast processing systems. Multiplication operation consumes high power and more hardware resources when compared to other arithmetic operations. This paper highlights the design and FPGA implementation of 4-bit Binary Vedic multiplier, which results in less power dissipation and low propagation delay when contrasted with conventional type of multipliers. The simulation of the vedic multipliers is carried out using Xilinx 2018 software with the HDL Verilog. The design is implemented and synthesized using FPGA board (Zed board). Also, the output is visualized using the board The performance analysis is conducted for the Vedic Binary multiplier designed with various adder architectures.**

Keywords*— *Vedic Mathematics, Ripple Carry Adder, Carry look ahead adder, Carry save adder, Array Multiplier

I. INTRODUCTION

The name "Vedic," which signifies the repository of all knowledge, is acquired from the Sanskrit word called "Veda", which implies a collection of sacred Hindu scriptures. An ancient mathematical method called Vedic Mathematics was used in India [1]. The basic arithmetic techniques are used to perform numerical calculations and solve problems in everyday life, science, engineering, and many other fields. Vedic mathematics employs a different method that is primarily based on 16 sutras [2]. These 16 sutras form the foundation of Vedic mathematics (also called as mathematical formulas or Vedas). Between AD 1911-1918, an Indian mathematician named Jagadguru Shri Bharathi Krishna Tirthaji made the discovery. He was revered as the founding father of Vedic mathematics [3]. It provides a simpler and more efficient approach to mathematics, and is particularly useful for mental calculations and problem-solving. The importance of Vedic mathematics lies in its ability to simplify complex calculations, reduce the time and effort required to solve problems, and enhance mental agility and sharpness. It provides a range of techniques that can be applied to a wide variety of mathematical problems, from simple addition and subtraction to more advanced topics such as algebra and calculus [4]. This fascinating area of study includes extremely powerful algorithms that may be used in many areas of engineering, including high speed computing and signal processing applications. The arithmetic operation, Multiplication could be integrated with methods of Vedic architectures to speed up the computation. The most crucial component in digital signal processing and other applications today is regarded as the multiplier [5]. Several academicians have attempted to design multipliers that either compromises on one of design goals: low power consumption, less space or high speed [6,7]. Vedic sutras can be employed to design high speed multipliers in various applications in Cryptography [8]. One among the 16 sutras is called the Urdhva-Triagbhyam meaning "Vertical and cross-wise" [9]. Using urdhva-triyagbhyam, alternative approaches to developing Vedic multipliers have been presented in this work.

The Vedic sutra "Urdhva Traigbhyam" refers to vertical and crosswise multiplication [10]. This approach is typically applied to decimal number systems. Comparing this technique to other multipliers, one benefit is that the delay and area rises gradually as the number of bits grow.

II. VEDIC MULTIPLIER

The Vedic multiplier is a specific technique within Vedic mathematics that can be implemented for the efficient multiplication. Vedic Multiplier is built on specific formulas and patterns derived from Vedic mathematics [11].

A. General structure

The multiplication using Vedic multiplier can be split into several steps [12]. At each step, sum of products is calculated, as depicted in Fig 1 (ie. resulting sum of step 1 is [a0*b0], resulting sum of step 2 is [a1*b0] + [b1*a0], resulting sum of step 3 is [b0*a2] + [b1*a1] + [b2*a0] and similarly for forth coming steps).

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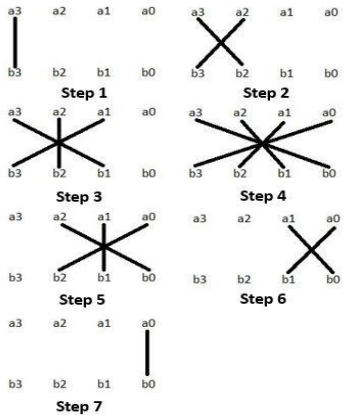


Fig 1: Basic Ideology behind 4x4 decimal multiplier using Urdhva Traigbhyam

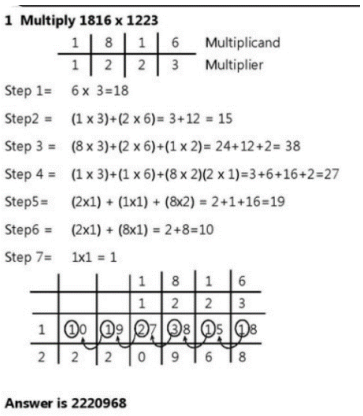


Fig 2: Example for 4x4 decimal multiplier using Vedic mathematics

From the above Fig 2, we can infer the final computation of Urdhva-Traigbhyam [13]. Resulting sums are arranged from right to left and their ones place is added to the tens place of the previous bit and noted down [5].

B. 2 x 2 Binary Vedic Multiplier

Consider A=a1a0 and B=b1b0 where a0 and b0 are LSB and a1 and b1 are MSB as shown in Fig 3. Now initialize s0 = a0*b0, c1s1 = a0*b1 + a1*b0, c2s2 = [c1 + a1*b1]. For step 1, we don't initialize carry because it has only one process (ie. Vertical Multiplication) whereas step 2 has two multiplications (cross-wise) and thus it is clear that step 2 produces carry (ie. c1). Now [c2 s2 s1 s0] are the resulting bits [14].

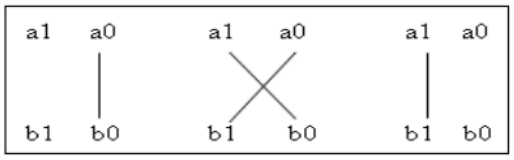


Fig 3: Vedic multiplication of 2x2 Binary multiplier

This method can be implemented using two Half Adders and four AND gates as depicted in Fig. 4.

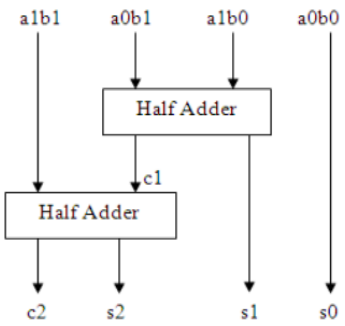


Fig 4: Structural Outline for 2x2 Binary Multiplier

C. 4 x 4 Vedic Multiplier

Let us consider A=[a3a2a1a0] & B=[b3b2b1b0], where A is considered as the multiplicand and B is considered as the multiplier. Now divide the formats of A & B into two equal parts so that for A –> {a3a2} & {a1a0}, for B –> {b3b2} & {b1b0}. Using the Vedic multiplication, we can take 2 bits at a time and perform 2x2 multiplication with the other 2 bits ie. 1st block as {a1a0 & b1b0}, last block as {a3a2 & b3b2} and the middle block has two blocks {a3a2 &b1b0} and {a1a0 & b3b2} as depicted in Fig 5.

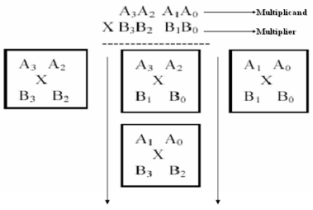


Fig 5: Architecture of 4x4 Vedic multiplication

So, the end result for this method of multiplication is 8-bit – [s7s6s5s4s3s2s1s0].

This implementation can be carried out with many adders and functions [15]. For example, Carry look ahead adder or Carry save adder or Ripple carry adder can be used.

In this paper, implementation of Vedic 4x4 Binary Multiplier has been carried out using different adder architectures on the Zed board.

III. VEDIC 4X4 MULTIPLIER USING VARIOUS ADDER ARCHITECTURES

Various adder architectures are used with 2x2 Vedic multiplier. The partial intermediate products are produced from 2x2 multiplier and the partial intermediate products are summed up to get the end results using various adder architectures.

A. Using Ripple Carry Addition architecture

Ripple Carry Adder is also quoted as parallel adder, the simplest topology for an adder structure. This adder structure is built by adding 'n' number of full adders serially. The architecture is such that each full adder consumes a carry input derived from the precedent full adder and produces carry output and sum. The carry output, thus produced is then provided as the input to the adjacent full adder.

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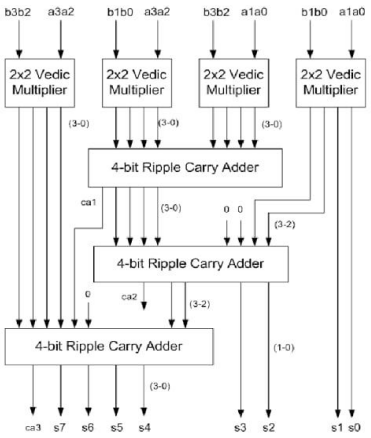


Fig 6: Structural outline of Vedic 4x4 Binary multiplier with ripple carry adder

In our design, four 2x2 Vedic multipliers and 3 ripple carry addition architectures are incorporated. Partial products are the outputs of the four 2x2 Vedic multipliers. The order of input bits is shown in the Structural Outline in Fig 6. The output [s0, s1] are directly obtained from 2x2 Vedic multiplier and rest of the outputs are obtained from the adders. While the last ripple carry adder gives 4 output bits [s4, s5, s6, s7] and remaining 2 output bits [s3, s2] are obtained from the other ripple carry adder. Step-by-step process of getting the output is clearly projected in the Structural Outline. In Verilog, a single module is used for ripple carry adder based on structural or gate level modelling.

B. Using Carry Look-ahead Addition architecture

The primary advantage of this adder is that it reduces carry propagation delay of adders. It involves the advance calculation of carry signals, depending on their respective inputs. The carry-in signals are generated using the input for each full adders and thus it makes sure that the any full adder does not wait for the carry-in.

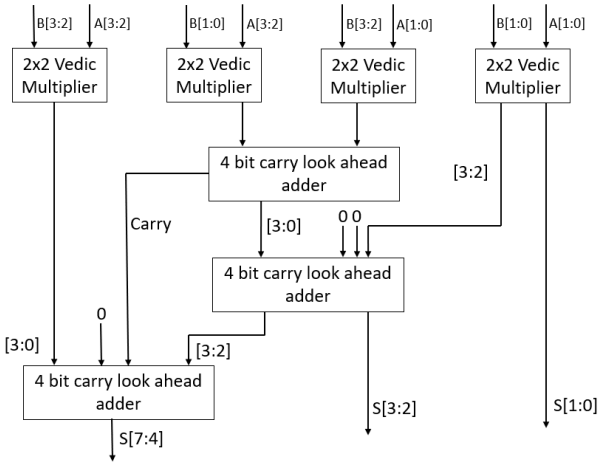


Fig 7: Structural Outline of Vedic 4x4 Binary Multiplier using Carry look-ahead adder

We use four 2x2 Vedic multipliers along with three carry look ahead adders. Partial products are outputs of the four 2x2 Vedic multipliers. The order of input bits is shown in the above Structural Outline. The output [s1, s0] are directly obtained from 2x2 Vedic multiplier and rest of the output are obtained from the adders. While the last carry look ahead adder gives 4 output bits [s4, s5, s6, s7] and remaining 2 output bits [s2, s3] are obtained from the other carry look ahead adder. Step-by-step process of getting the output is clearly projected in the Structural Outline in Fig 7. In Verilog, a single module can be employed for implementing a look ahead addition architecture with any modelling style: structural or gate level modelling. The working of Vedic 4x4 binary multiplier using look ahead addition is more or less similar to the working only with ripple carry addition architecture in it. So, replacing all the ripple carry adders and incorporating the new look ahead addition architecture gives the same output (i.e. the end output of the Vedic 4x4 multiplier).

C. Using Carry Save Addition Architecture

Carry Save Addition can be employed to efficiently calculate the sum of binary numbers. As multiplication involves repeated additions, this adder structure can be used to compute the end result, involving the addition of the intermediate partial products. This adder architecture is proven to be faster than other architectures, wherein the propagation delay can be minimized by employing carry save addition. The idea behind the addition of three numbers is that the carry information has to be passed for every corresponding addition; but in carry save adder, the carry is calculated initially for all the set of addition and added to the sum until the very last step.

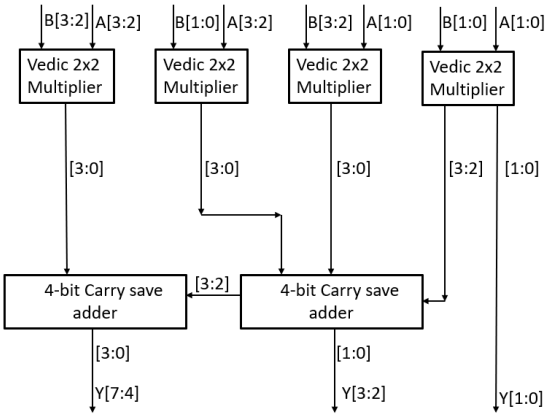


Fig 8: Structural Outline of Vedic 4x4 Binary Multiplier using Carry Save Adder

We use four 2x2 Vedic multipliers and two carry save adders. Partial products are the outputs of the four 2x2 Vedic multipliers. The order of input bits is shown in the Structural Outline in Fig 8. The output Y(1:0) are directly obtained from 2x2 Vedic multiplier and rest Y(7:2) are obtained from the adders. We have two carry save adders that obtains Y(7:4) and Y(3:2) as the output. Step-by-step process of getting the output is clearly projected in the Structural Outline. In Verilog, we can use a single module for carry save adder based on structural or gate level modelling.

IV. COMPARISON OF VARIOUS ADDER ARCHITECTURES

The Verilog programs are simulated using Xilinx Vivado and the multiplier architectures are implemented on the FPGA board. In this section, a comparison of the power consumption and number of LUTs used for various adder architectures is presented in Fig 9 to Fig 14.

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A. *Ripple carry adder*

The Fig 9 illustrates the power consumed by a Vedic binary multiplier implemented with the help of a Ripple Carry Addition architecture. The number of LUTs used in the implementation using the above logic is illustrated in Fig 10.

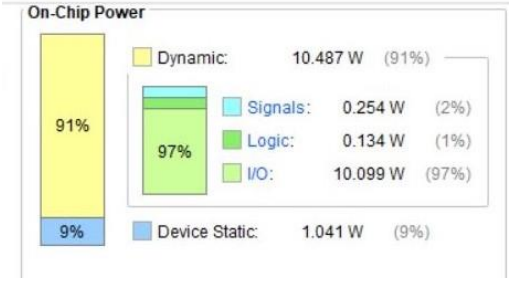


Fig 9: Power consumption of the Vedic Binary multiplier created with Ripple Carry Addition architecture

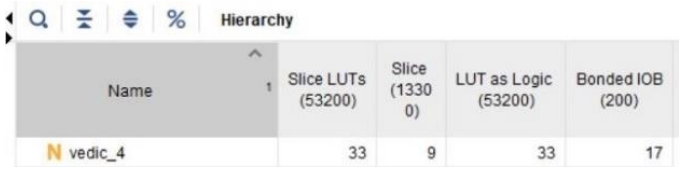


Fig 10: Number of LUTs used for the Vedic multiplier created using Ripple Carry Addition architecture

B. *Carry-Look Ahead Adder*

Fig 11 illustrates power consumed by a Vedic binary multiplier implemented with the help of a Carry Look Ahead Addition architecture. Fig 12 represents the number of LUTs used in the implementation using the above logic.

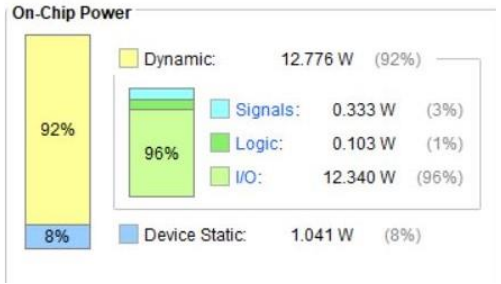


Fig 11: Power consumption for the Vedic Binary multiplier created with Carry Look Ahead Addition architecture

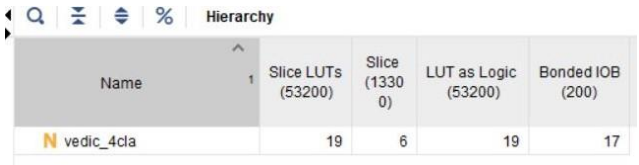


Fig 12: Number of LUTs used for Vedic Binary multiplier created with Carry Look Ahead Addition Architecture

C. *Carry Save Adder*

Fig 13 illustrates power consumed by a Vedic binary multiplier implemented using a Carry Look Ahead Addition architecture. Fig 14 represents the number of LUTs used in the implementation using the above logic.



Fig 13: Power consumption for Carry Save Adder Architecture



Fig 14: Number of LUTs used for Carry Save Adder Architecture

TABLE 1: COMPARISON OF THE ON-CHIP POWER AND NUMBER OF LUTS FOR VARIOUS ADDER ARCHITECTURES

Types of Adder Architectures	Power consumption	LUTs used
Ripple Carry Adder	11.528 W	33
Carry Look Ahead Adder	13.817 W	19
Carry Save Adder	12.853 W	21

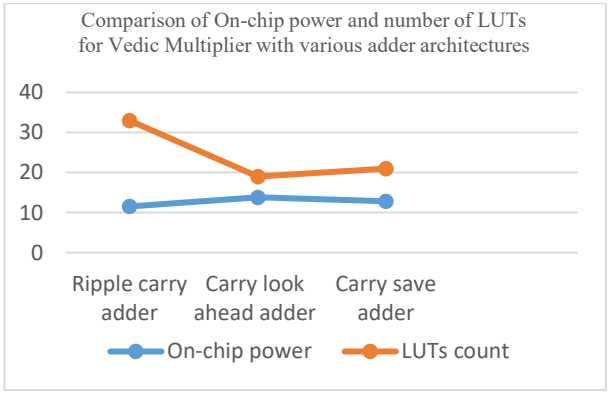


Fig 15: Comparison of On-chip power and number of LUTs for Vedic Multiplier with various adder architectures

The snapshot of the FPGA board (Zed board) is attached below that shows the output when two 4 bit numbers are multiplied (i.e. 0001 x 0111 = 00000111)

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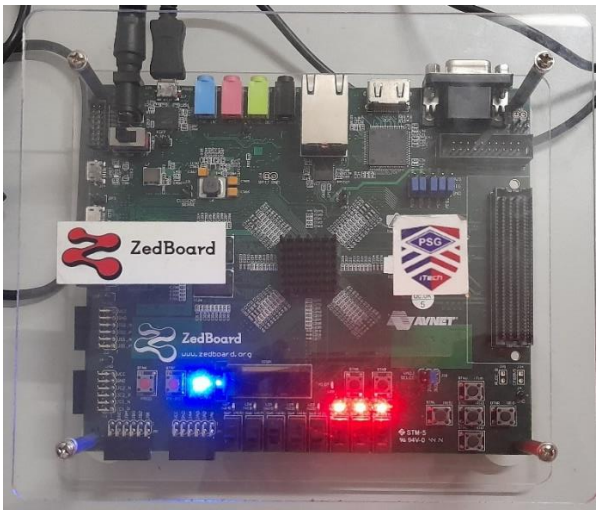


Fig 16: Zed board Evaluation and Development kit – Output of Vedic 4x4 Binary multiplier displayed using the LEDs

V. CONCLUSION

The Verilog codes were written for Vedic 4x4 Multiplier using various adder architectures in Xilinx Vivado. The on-chip power consumption comprising of static and dynamic power along with the number of LUTs used for all three architectures were presented and it was inferred that there exists a tradeoff between the power consumed and the number of LUTs, during the implementation in the Zedboard. The power and the LUTs used were found to be lower than the conventional implementation of multipliers.

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